

Bluestock Mutual Fund Analytics Capstone

Final Project Report

1. Project Overview

An end-to-end mutual fund analytics project covering data ingestion, ETL, SQLite database creation, exploratory data analysis, performance measurement, risk analytics, investor analytics, recommendations and interactive dashboarding.

2. Dataset

The project uses 10 datasets covering fund master information, daily NAV history, AUM, SIP inflows, category inflows, folio counts, scheme performance, investor transactions, portfolio holdings and benchmark indices.

3. Data Engineering

The ETL pipeline validates the source data, cleans NAV observations, loads the datasets into SQLite and creates the analytical tables required for downstream analysis.

4. Performance Analytics

Fund performance is evaluated using CAGR, annualised return, annualised volatility, Sharpe ratio, beta, historical Value at Risk and maximum drawdown.

5. Risk Analytics

Risk is analysed using volatility, Sharpe ratio, beta, historical VaR and maximum drawdown. Risk-versus-return comparisons help identify funds offering stronger return characteristics relative to their observed risk.

6. Investor Analytics

Investor transactions are analysed by age group, city tier, payment mode, transaction type, gender, state and city to identify behavioural patterns.

7. Fund Recommendations

A scoring framework combines three-year return, Sharpe ratio, expense ratio and maximum drawdown to rank funds.

8. Dashboard

The proposed Power BI dashboard contains executive overview, fund performance, risk analytics, investor analytics, and market/portfolio analytics pages with interactive slicers.

9. Conclusion

The project provides a complete analytical workflow from raw mutual fund data to actionable performance, risk and investor insights. The resulting database, notebooks, recommendation engine and dashboard specification provide a reproducible foundation for mutual fund analysis.

Key Formulas

Metric	Formula
CAGR	$(\text{Ending NAV} / \text{Beginning NAV})^{(252 / n)} - 1$
Sharpe	$\text{Annualised Return} / \text{Annualised Volatility}$
Beta	$\text{Covariance}(\text{Fund Return}, \text{Benchmark Return}) / \text{Variance}(\text{Benchmark Return})$
Historical VaR 95%	-5th percentile of daily returns
Maximum Drawdown	$\text{Minimum}((\text{Wealth} / \text{Running Maximum}) - 1)$